

AAI Vadodara Airport — IT Asset Manager v2.0

Complete System Features, Database Columns & Business Logic Specification

Location: Vadodara Airport (VABO) | Environment: Enterprise | Date: September 10, 2026

1. Complete Features List (Site Capabilities & Modules)

■ 1. Dashboard & Real-Time Analytics

- Live counters for Total Assets, In Stock (Store), and Issued (Assigned) items.
- Dynamic category distribution visual chart with interactive legends.
- Real-time Recent Activity feed tracking latest handovers and registrations.
- System Diagnostics & Health monitoring endpoints (/api/health, /api/info).

■ 2. Asset Inventory & Data Controls

- Fast client & server paginated inventory table (default 50 items/page).
- Multi-attribute search filtering by Name, Serial, Holder, Department, IP, and Hostname.
- Type selector grid (Filter by PC, AIO, Laptop, Printer, UPS, Scanner, Xerox, etc.).
- Department selector grid for viewing assets assigned to specific airport units.

🏠 ■ 3. Single Asset Registration & Auto Tagging

- Auto-generates enterprise Asset Tags (format: VABO-IT/CNS-{DEPT}-{TYPE}-{SEQ}).
- Dynamic input forms revealing specific fields based on hardware selection (Monitor Make/Serial, Charger Serial, KVA rating, etc.).
- Dual-holder tracking: Current System Holder vs Physical Asset Holder (contractor).
- Serial number uniqueness validation prior to database entry.

■■■■ 4. Unified Multi-Asset Entry Wizard

- 5-step wizard to assign multiple bundled assets to an employee in one workflow.
- Step 1: User Info with employee search auto-complete.
- Step 2: Hardware asset categories multi-selection grid.
- Step 3: Technical specifications and serial numbers input form.
- Step 4: Shared network context (IP Address & Hostname).
- Step 5: Review & Save with 100% single-transaction SQL integrity (BEGIN...COMMIT/ROLLBACK).

■ 5. Handover & Takeover Workflows

- Handover: Issue store hardware items to staff with IP/Hostname assignment.
- Takeover: Receive items back to IT Store, auto-clearing employee data and network bindings.
- Auto-generates reference vouchers (AAI/VABO/IT/{TYPE}/{YEAR}/{RAND}).
- Printable receipt view formatted for A4 printing and PDF generation.

■ 6. Mutual Asset Linking & Unlinking

- Parent-Child mutual asset pairing (e.g., PC tower linked to Monitor, or Printer to Laptop).
- Mutual pointer enforcement (parent.linked_asset_id → child.id).
- One-click unlinking operation with dedicated link/unlink audit transactions.

■ 7. Audit Logs & Revision Tracking

- Complete transaction history filtered by type (handover, takeover, link, unlink).
- Admin Audit Edit mode with full change diff tracking in edit_history JSON log.
- Soft-delete system with mandatory reason logging (is_deleted = 1).
- Record protection flag (is_protected = 1) to block accidental deletion.

■ 8. Document Print Security Audit Logs

- Automatic print logging for every receipt printed or reprinted (POST /api/transactions/print-logs).
- Logs printed-by username, user department, client IP address, and client hostname.
- Tracks reference numbers and document types printed.

■ 9. Employee Directory & Asset Trail

- Full employee directory (Name, Designation, Department, Active/Inactive status).
- Lifetime Asset Trail timeline modal showing all historical items assigned to an employee.
- Multi-criteria sorting (Name, Department, Designation, Purchase Year).

■ 10. Accounts & Access Control Manager (Port 8002)

- Independent admin web application running on http://localhost:8002.
- User creation and role management (admin vs user).
- Bcrypt password reset functionality and JWT token session management (24h expiry).

■ 11. Bulk Import (UPSERT), Export & Backups

- Bulk CSV Import with UPSERT (ON CONFLICT(serial_number) DO UPDATE).
- Import preview table verifying successful vs failed rows with error reporting.
- Multi-table CSV export for Assets, Employees, Transactions, Print Logs, and Schema.
- JSON Database backup and snapshot restoration.

■ 12. Barcode Scanning & UI Personalization

- Integrated camera Barcode / QR Code reader (html5-qrcode).
- Theme toggle (Light Mode vs Glassmorphic Dark Mode) and font scaling (A-, A, A+).
- Cross-platform deployment (Web SPA, Termux Android, Docker, PyQt5 Desktop).

2. Complete Database Schema & Columns List

1. Table: users (System Accounts - 4 Columns)

Column Name	Data Type	Constraints	Description
id	SERIAL / INT	PRIMARY KEY	Unique user account ID
username	VARCHAR(255)	NOT NULL, UNIQUE	Account login username
password	VARCHAR(255)	NOT NULL	Bcrypt password hash (salt=10)
role	VARCHAR(50)	DEFAULT 'user'	Role (admin or user)

2. Table: employees (Staff Directory - 6 Columns)

Column Name	Data Type	Constraints	Description
id	SERIAL / INT	PRIMARY KEY	Unique employee ID
name	VARCHAR(255)	NOT NULL, UNIQUE	Full employee name
designation	VARCHAR(255)	DEFAULT ''	Job title / designation
department	VARCHAR(255)	DEFAULT ''	Assigned department
status	VARCHAR(50)	DEFAULT 'Active'	Status (Active / Inactive)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date profile created

3. Table: assets (Hardware Inventory - 25 Columns)

Column Name	Data Type	Constraints	Description
id	SERIAL / INT	PRIMARY KEY	Unique asset ID
name	VARCHAR(255)	NOT NULL	Asset category (PC, Laptop, Printer, etc.)
serial_number	VARCHAR(255)	NOT NULL, UNIQUE	Hardware serial number (upsert key)
asset_tag	VARCHAR(255)	DEFAULT ''	Asset tag (VABO-IT/CNS-DEPT-TYPE-NO)
charger_serial	VARCHAR(255)	DEFAULT ''	Laptop power adapter serial number
monitor_make	VARCHAR(255)	DEFAULT ''	Monitor manufacturer make
monitor_serial	VARCHAR(255)	DEFAULT ''	Monitor serial number
keyboard_make	VARCHAR(255)	DEFAULT ''	Keyboard manufacturer make
mouse_make	VARCHAR(255)	DEFAULT ''	Mouse manufacturer make
make	VARCHAR(255)	DEFAULT ''	Main asset make (HP, Dell, Lenovo)
model	VARCHAR(255)	DEFAULT ''	Hardware model name/number
ip_address	VARCHAR(255)	DEFAULT ''	Assigned network IP address
hostname	VARCHAR(255)	DEFAULT ''	Assigned domain hostname
current_user	VARCHAR(255)	DEFAULT 'IT Store'	System asset holder ('IT Store' or Name)
contractual_user_name	VARCHAR(255)	DEFAULT ''	Physical asset holder / contractor
assigned_dept	VARCHAR(255)	DEFAULT ''	Department assigned to
assigned_desig	VARCHAR(255)	DEFAULT ''	Designation assigned to
employee_id	INTEGER	FK → employees(id)	Assigned employee foreign key
linked_asset_id	INTEGER	FK → assets(id)	Linked paired asset foreign key
status	VARCHAR(255)	DEFAULT 'In Stock'	Status ('In Stock' / 'Assigned')
remark	TEXT	DEFAULT ''	Asset notes or condition state
year_of_purchase	INTEGER	DEFAULT NULL	Purchase year (YYYY)
kva	VARCHAR(255)	DEFAULT ''	Power rating (for UPS units)
warranty_expiry	VARCHAR(255)	DEFAULT ''	Warranty expiry date string
last_update	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of last edit

4. Table: transactions (Audit & Movement Logs - 24 Columns)

Column Name	Data Type	Constraints	Description
id	SERIAL / INT	PRIMARY KEY	Unique transaction ID
type	VARCHAR(50)	NOT NULL	Type (handover, takeover, link, unlink)
asset_ids	TEXT	NOT NULL	JSON array string of asset IDs
asset_names	TEXT	DEFAULT ''	Comma-separated asset descriptions
ref_no	VARCHAR(255)	DEFAULT ''	Reference code (AAI/VABO/IT/...)
date	VARCHAR(255)	DEFAULT ''	Transaction execution date
employee_name	VARCHAR(255)	NOT NULL	Target employee name
employee_desig	VARCHAR(255)	DEFAULT ''	Target employee designation
employee_dept	VARCHAR(255)	DEFAULT ''	Target employee department
issuer_name	VARCHAR(255)	DEFAULT ''	Issuing officer name
issuer_desig	VARCHAR(255)	DEFAULT ''	Issuing officer designation
issuer_dept	VARCHAR(255)	DEFAULT ''	Issuing officer department
employee_id	INTEGER	FK → employees(id)	Target employee foreign key
issuer_id	INTEGER	FK → employees(id)	Issuer employee foreign key
remark	TEXT	DEFAULT ''	Transaction remark / notes
edit_history	TEXT	DEFAULT '[]'	JSON array of admin revision diffs
last_edited_by	VARCHAR(255)	DEFAULT ''	Username of last admin editor
last_edited_at	TIMESTAMP	DEFAULT NULL	Timestamp of last admin edit
timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
is_deleted	INTEGER	DEFAULT 0	Soft delete flag (0 active, 1 deleted)
deleted_by	VARCHAR(255)	DEFAULT ''	Admin username who deleted log
deleted_at	TIMESTAMP	DEFAULT NULL	Deletion timestamp
delete_reason	TEXT	DEFAULT ''	Reason for record deletion
is_protected	INTEGER	DEFAULT 0	Protection flag (1 prevents deletion)

5. Table: print_logs (Print Security Audit - 10 Columns)

Column Name	Data Type	Constraints	Description
id	SERIAL / INT	PRIMARY KEY	Unique print log ID
transaction_id	INTEGER	DEFAULT NULL	Associated transaction ID
action_type	VARCHAR(50)	DEFAULT 'Print'	Action (Print / Reprint)
printed_by	VARCHAR(255)	NOT NULL DEFAULT ''	Username who printed voucher
printed_by_dept	VARCHAR(255)	DEFAULT ''	Department of user printing
system_ip	VARCHAR(255)	DEFAULT ''	Client computer IP address
system_hostname	VARCHAR(255)	DEFAULT ''	Client computer hostname
ref_no	VARCHAR(255)	DEFAULT ''	Reference code printed
doc_type	VARCHAR(255)	DEFAULT ''	Document type (Handover, Takeover)
print_timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Print timestamp

3. Complete Business Logic & Workflow Rules

1. Authentication, Password Hashing & JWT Security

Passwords are standardly encrypted using Bcrypt with 10 salt rounds before database storage. API authorization uses JWT (JSON Web Tokens) with a 24-hour expiry window. Admin-only endpoints enforce `req.user.role === 'admin'` before executing sensitive updates or soft deletions. Default admin account (admin / admin123) is seeded automatically if the user database table is empty.

2. Automatic Asset Status Calculation

Asset status is determined dynamically based on the current holder string: `const status = (currentUser.trim().toLowerCase() === 'it store') ? 'In Stock' : 'Assigned';`. If holder is 'IT Store', status automatically becomes 'In Stock'. If assigned to any employee, status becomes 'Assigned'. System maintains separate fields for official holder (`current_user`) and physical contractual user (`contractual_user_name`).

3. Asset Tag Auto-Generation Sequence Rule

Asset tags follow the enterprise formula: VABO-IT/CNS-{DEPT}-{TYPE}-{SEQ}. The sequence engine queries existing tags with the prefix, parses the highest integer suffix, increments by 1, and formats the tag padded to 2 digits (e.g. VABO-IT/CNS-IT-PC-04).

4. Handover State Mutation Workflow

When issuing hardware (type = 'handover'): 1) Inserts audit record into transactions table with reference code AAI/VABO/IT/HANDOVER/... 2) Updates each asset: `status = 'Assigned'`, `current_user = employee_name`, `assigned_dept = employee_dept`, `assigned_desig = employee_desig`, `ip_address = new_ip`, and `hostname = new_hostname`.

5. Takeover State Mutation & Network Cleanup

When receiving hardware back into store (type = 'takeover'): 1) Inserts audit record into transactions table with reference code AAI/VABO/IT/TAKEOVER/... 2) Resets each asset: `status = 'In Stock'`, `current_user = 'IT Store'`, clears department/designation, sets `employee_id = NULL`, and wipes network bindings (`ip_address = ''`, `hostname = ''`, `contractual_user_name = ''`).

6. Unified Entry Wizard Transactional Integrity

Bundling multiple assets to an employee runs within a single SQL transaction block (BEGIN TRANSACTION ... COMMIT / ROLLBACK): 1) Checks input serial numbers for duplicates against existing DB. If duplicate found, aborts with 409 Conflict. 2) Syncs employee profile in employees table. 3) Inserts all asset rows, creates transaction record, and commits. If any single asset insert fails, entire batch rolls back with 0 side effects.

7. Mutual Parent-Child Asset Linking Algorithm

Establishing asset pairs (e.g. PC Tower and Monitor): 1) Clears existing pointers on both records (`linked_asset_id = NULL`). 2) Sets reciprocal pointers: `parent.linked_asset_id = child.id` AND `child.linked_asset_id = parent.id`. 3) Logs dedicated transaction record (type = 'link' or 'unlink').

8. Admin Audit Revision & History Diff Tracking

Modifying audit records requires admin permission. The engine compares modified fields against original database values, extracts changed keys into a structured JSON diff object: `{ edited_by: 'admin', edited_at: ISO_DATE, changes: { employee_name: { old: 'John', new: 'John Doe' } } }`. Appends this diff to the record's `edit_history` array for permanent audit transparency.

9. Bulk CSV Import UPSERT Strategy

Bulk asset CSV import uses SQL `ON CONFLICT(serial_number) DO UPDATE`. Existing assets matching by serial number have their details updated in place, while new serial numbers are inserted seamlessly without duplicate key crash exceptions.

10. PostgreSQL Cloud & SQLite Hybrid Translation Layer

`config/db.js` acts as a hybrid database adapter. It auto-translates SQLite syntax (`?` placeholders, `datetime('now')`, `LIKE`) into Neon PostgreSQL syntax (`$1`, `$2...`, `CURRENT_TIMESTAMP`, `ILIKE`, `RETURNING id`), enabling seamless operation on both local SQLite file databases and Neon PostgreSQL cloud clusters.